



MATEMATICI 2

Curs 16

Teorema lui Liauville

Dacă $W(t) = \det(X(t))$ = wronskianul sist. fund. de sol. $x^{(1)}, \dots, x^{(n)}$ $\left(X(t) = (x^{(1)}(t) \mid \dot{x}^{(2)}(t) \mid \dots \mid x^{(n)}(t)\right)$

atunci $W(t) = W(t_0) \cdot e^{\int_{t_0}^{t} \mathrm{Tr}(A(\tau))d\tau}$.

$W(t_0) \neq 0 \Rightarrow$ sistem de sol. fundamental.

Teoremă : Fie sist. $x'(t) = A(t)x(t) + b(t)$, $x(t) = \begin{pmatrix} x_1(t) \\ \vdots \\ x_n(t) \end{pmatrix}$

$A(t) = (a_{ij}(t))_{1 \le j,i \le n}$, $b(t) = \begin{pmatrix} b_1(t) \\ \vdots \\ b_n(t) \end{pmatrix}$

Notăm cu $S = \{ x : I \to \mathbb{R}^n \mid x'(t) = A(t)x(t) \}$

Dacă x_p e o sol. particulară a sist.

(1) și $\mathcal{S}$ = mult. sol. sist. (1), at. $\mathcal{S} = x_p + S = \{ x_p + x \mid x \in S \}$

Teorema lui Lagrange

Fie pb. Cauchy $x' = A(t)x + b(t)$

$x(t_0) = x_0$, $t_0 \in I$, $x : I \to \mathbb{R}^n$

Dacă $X(t)$ = o matrice fundamentală a sist. $x' = A(t) \cdot x$,

soluția pb. Cauchy este $x(t) = (X(t))\left(X^{-1}(t_0)\, x_0 + \int_{t_0}^{t} X^{-1}(\tau) b(\tau) d\tau\right)$

Dem: $x_0(t) = x(t) \cdot c$

Căutăm o sol. particulară prin metoda var. ct. (Lagrange)

de forma $x_p(t) = x(t) \cdot c(t)$.

În final $x(t) = x_0(t) + x_p(t) = X(t) \cdot C + X(t) \int_{t_0}^{t} X^{-1}(\tau) f(\tau)\, d\tau$.

$x(t_0) = x_0 \Rightarrow C = X^{-1}(t_0) \cdot x_0$.

Rezolvarea sistemelor de ecuații cu coeficienți constanți :

- în cazul în care $A(t)$ - matrice cu., sist. prezentate ant., s.n. sist. cu coef. ct. $t \in I$ Astfel, aceeași, $\underline{x' = A \cdot x}$, cu $A \in \mathcal{M}_n(\mathbb{R})$

s.n. sistem omogen → pb. Cauchy este det. unei sol. $\{x \mid x(t_0) = x_0, t_0 \in I\}$. De asemenea, $x' = A \cdot x + b(t)$, $t \in I \subset \mathbb{R}$ cu pb. Cauchy $x(t_0) = x_0 \in \mathbb{R}^n$, $t_0 \in I$.

Propr. : Dacă $\lambda_1 \in \sigma(A)$ = mult. real. proprii ale lui A | spectru și $u_1 \in V_{\lambda_1}$, at.: $X^{(1)}(t) = e^{\lambda_1 t} \cdot u_1$ este sol. a sist. neomogen.

Dem. : $(e^{\lambda_1 \cdot t} \cdot u_1)' = \lambda_1 e^{\lambda_1 t} u_1 = e^{\lambda_1 t} \cdot \lambda_1 u_1 = e^{\lambda_1 t} A \cdot u_1 =$

$= A \cdot (e^{\lambda_1 t} \cdot u_1)$

Consecință : Dacă A este diagonalizabilă și are val. proprii $\{\lambda_1, \dots \lambda_n\}$ cu vect. proprii $\{u_1, \dots, u_n\}$ at. sist. de sol. $\{x^{(1)}(t) = e^{\lambda_1 t} u_1, \dots, x^{(n)}(t) = e^{\lambda_n t} u_n\} = sist.$ fundamental de soluții.

Fie $\alpha_1 x^{(1)} + \dots + \alpha_n x^{(n)} = 0$

$\alpha_1 x^{(1)}(t_0) + \dots + \alpha_n x^{(n)}(t_0) = 0(t_0)$

$\alpha_1 e^{\lambda_1 t_0} u_1 + \dots + \alpha_n e^{\lambda_n t_0} u_n = 0 \Rightarrow$

$\Rightarrow \alpha_1 e^{\lambda_1 t_0} = \dots = \alpha_n e^{\lambda_n t_0} = 0 \Rightarrow \underline{\alpha_1 = \dots = \alpha_n = 0}$

exemple: Să se det. sol. generală a sistemului:

$$\begin{cases} y_1' = 3y_1 - y_2 + y_3 \\ y_2' = -y_1 + 5y_2 - y_3 \\ y_3' = y_1 - y_2 + 3y_3 \end{cases}$$

Se cere sol. generală și sol. care verifică $y_1(0) = 1$, $y_2(0) = 1$, $y_3(0) =$

$$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}' = \begin{pmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{pmatrix} \cdot \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}; \quad A = \begin{pmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{pmatrix}$$

$$Y = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} \Rightarrow$$

$$\Rightarrow Y' = A \cdot Y$$

$$\det(A - \lambda I_3) = 0 \Rightarrow \begin{vmatrix} 3-\lambda & -1 & 1 \\ -1 & 5-\lambda & -1 \\ 1 & -1 & 3-\lambda \end{vmatrix} = 0 \Rightarrow$$

$$\xrightarrow{l_1 - l_3} \begin{vmatrix} 2-\lambda & 0 & -2+\lambda \\ -1 & 5-\lambda & -1 \\ 1 & -1 & 3-\lambda \end{vmatrix} = (2-\lambda) \cdot \begin{vmatrix} 1 & 0 & -1 \\ -1 & 5-\lambda & -1 \\ 1 & -1 & 3-\lambda \end{vmatrix} =$$

$$= (2-\lambda) \cdot \begin{vmatrix} -1 & -1 \\ 1 & 3-\lambda \end{vmatrix} \cdot (-1)^3 = (\lambda-2)(\lambda-3+1) =$$

$$= (\lambda-2)^2 = 0 \quad \ldots \quad (\text{Temă: } \lambda_1 = 2;\ \lambda_2 = 3;\ \lambda_3 = 6)$$

$$V_{\lambda_1} = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 \mid (A - \lambda_1 I_3) \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \right\}$$

$$\lambda_1: \begin{pmatrix} 1 & -1 & 1 \\ -1 & 3 & -1 \\ 1 & -1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow \begin{cases} x - y + z = 0 \\ -x + 3y - z = 0 \\ x - y + z = 0 \end{cases}$$

$$2y = 0 \Rightarrow y = 0$$

$$\Rightarrow x = -z \Rightarrow V = \{ (\alpha, 0, -\alpha) \}.$$

$$\text{pt. } \alpha = 1 \Rightarrow V = \{ \alpha(1, 0, -1) \}.$$

pt. $x = \alpha \Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -\alpha \\ 0 \\ \alpha \end{pmatrix} = \alpha \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$

$u_1 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$, $Y^{(1)}(t) = e^{2x} \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$.

$\lambda_2: \begin{pmatrix} 0 & -1 & 1 \\ 1 & +2 & -1 \\ 1 & -1 & 0 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow \left.\begin{array}{l} z = \beta \\ x = \beta \\ y = \beta \end{array}\right\} \Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \beta \\ \beta \\ \beta \end{pmatrix} = \beta \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \Rightarrow$

$\Rightarrow Y^{(2)}(t) = e^{3x} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

λ_3: $\begin{pmatrix} -3 & -1 & 1 \\ 1 & -1 & -1 \\ 1 & -1 & -3 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow \begin{cases} z = \delta \\ -3x - y = -\delta \\ -x - y = \delta \end{cases} \Rightarrow$

$\Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \delta \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} \Rightarrow Y^{(3)}(t) = e^{6x} \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}.$

$$Y(x) = \begin{pmatrix} y_1(x) \\ y_2(x) \\ y_3(x) \end{pmatrix} = C_1 \cdot y^{(1)}(x) + C_2\, y^{(2)}(x) + C_3\, y^{(3)}(x) =$$

$$= \underbrace{\left(y^{(1)}(x) \mid y^{(2)}(x) \mid y^{(3)}(x) \right)}_{Y(x) = \text{matricea fundam.}} \begin{pmatrix} C_1 \\ C_2 \\ C_3 \end{pmatrix}$$

$$y_1(x) = c_1 \cdot e^{2x} \cdot (-1) + c_2 e^{3x} \cdot 1 + c_3 \cdot e^{6x} \cdot 1$$

$$y_2(x) = c_1 e^{2x} \cdot 0 + c_2 e^{3x} \cdot 1 + c_3 e^{6x} \cdot (-2) \quad ; c_1, c_2, c_3 \in \mathbb{R}$$

$$y_3(x) = c_1 e^{2x} \cdot 1 + c_2 e^{3x} \cdot 1 + c_3 e^{6x} \cdot 1$$

$$y_1(0) = 1$$

$$y_2(0) = 1$$

$$y_3(0) = 1$$

$$\begin{cases} -C_1 + C_2 + C_3 = 1 \\ C_1 \cdot 0 + C_2 - 2C_3 = 1 \\ C_1 + C_2 + C_3 = 1 \end{cases} \Rightarrow \boxed{C_1 = 0}$$

$$3C_3 = 0 \Rightarrow \boxed{C_3 = 0}$$

$$\boxed{C_2 = 1}$$

$\Rightarrow y(x) = 0 \cdot y^{(1)}(x) + 1 \cdot y^{(2)}(x) + 0 \cdot y^{(3)}(x)$ sol. prob. Cauchy

- în cazul în care sistemul este omogen, după det. lui $x(t)$ folosim ple. Cauchy (vezi th. lui Lagrange).

Să se integr. prin met. var. ct:

$$\begin{cases} y_1' = y_2 \\ y_2' = y_1 + e^x + e^{-x} \end{cases}$$

Rez: $\begin{cases} y_1' = y_2 \\ y_2' = y_1 \end{cases} \Rightarrow \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}' = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$

$$y' = A \cdot y.$$

$$\det(A-\lambda I_2)=0 \Rightarrow \begin{vmatrix} -\lambda & 1 \\ 1 & -\lambda \end{vmatrix}=0 \Rightarrow \lambda=\pm 1 \quad (\lambda^2-1=0)$$

$$\lambda_1=1 \Rightarrow V_{\lambda_1}=\left\{\begin{pmatrix} x \\ y \end{pmatrix}\in \mathbb{R}^2 \mid (A-\lambda_1 I_2)\cdot\begin{pmatrix} x \\ y \end{pmatrix}=\begin{pmatrix} 0 \\ 0 \end{pmatrix}\right\}$$

$$\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}\cdot\begin{pmatrix} x \\ y \end{pmatrix}=\begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow x=y=\alpha \Rightarrow \begin{pmatrix} x \\ y \end{pmatrix}=\alpha\begin{pmatrix} 1 \\ 1 \end{pmatrix} \Rightarrow$$

$$\Rightarrow y^{(1)}=e^{1\cdot x}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$V_{\lambda_2} = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 \;/\; (A - \lambda_2 I_2) \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\}$$

$$\Rightarrow \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow x = -y = \alpha \Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = \alpha \begin{pmatrix} 1 \\ -1 \end{pmatrix} \Rightarrow$$

$$\Rightarrow y^{(2)} = e^{-1 \cdot x} \begin{pmatrix} 1 \\ -1 \end{pmatrix} ; \; y = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

$$y_0(x) = \begin{pmatrix} y_{01}(x) \\ y_{02}(x) \end{pmatrix} = c_1 \cdot y_1(x) + c_2\, y_2(x) =$$
$$= c_1 \cdot e^{x}\begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2\, e^{-x}\begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Metoda 1 :

$$Y(x) = \left(y^{(1)}(x) \mid y^{(2)}(x)\right) = \begin{pmatrix} e^{x}\cdot 1 & e^{-x}\cdot 1 \\ e^{x}\cdot 1 & e^{-x}(-1) \end{pmatrix}$$

= matricea fundam. a sistemului.

$$y_0(x) = Y(x)\cdot\begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

$$y(x) = Y(x)\left(\begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \int_{t_0}^{t} Y^{-1}(\tau)\cdot b(\tau)\,d\tau\right),\ t_0 \in \mathbb{R}.$$

Metoda 2 : caut $y_p(x)$ de forma sol. ec. omog., punând în locul lui $c_1(x)$ și $c_2(x)$ $\Rightarrow y_p(x) = c_1(x) \cdot y^{(1)}(x) + c_2(x) y^{(2)}(x) \Rightarrow$

$$\Rightarrow \begin{cases} y_{1p}(x) = c_1(x) \cdot e^x \cdot 1 + c_2(x) e^{-x} \cdot 1 \\ y_{2p}(x) = c_1(x) e^x \cdot 1 + c_2(x) e^{-x} \cdot (-1) \end{cases}$$

$$\begin{cases} \left(C_1(x)e^{x}\cdot 1 + C_2(x)e^{-x}\cdot 1\right)' = C_1(x)e^{x} + C_2(x)e^{-x}(-1) \\ \left(C_2(x)e^{-x}(1)\cdot(-1) + C_1(x)e^{+x}\cdot 1\right)' = C_1(x)e^{+x}(+1) + C_2(x)e^{-x}(+1) \end{cases}$$

$$+e^{x}+e^{-x}$$

$$\begin{cases} C_1'(x)\,e^{x} + C_1(x)e^{x} + C_2'(x)\,e^{-x} - C_2(x)e^{-x} = C_1(x)e^{x} + \\ + C_2(x)e^{-x}(-1) \\ C_1'(x)\,e^{x} + C_1(x)e^{x} + C_2'(x)e^{-x}(-1) + C_2(x)e^{-x} = \end{cases}$$

$$= C_1(x)e^{x} + C_2(x)e^{-x} + e^{x} + e^{-x}$$

$$\Rightarrow \begin{cases} C_1'(x)e^{x} + C_2'(x)e^{-x} = 0 \\ C_1'(x)e^{x} + C_2'(x)e^{-x}(-1) = e^{x} + e^{-x} \end{cases}$$

$$2C_1'(x)e^{x} = e^{x} + e^{-x} \quad / \cdot \frac{e^{-x}}{2} \Rightarrow$$

$$\Rightarrow C_1'(x) = \frac{1}{2} + \frac{e^{-2x}}{2} \Rightarrow \boxed{C_1(x) = \frac{1}{2}x + \frac{(-e^{-2x})}{4}}$$

(-) $\Rightarrow 2C_2'(x)e^{-x} = -e^{x} - e^{-x} \quad / \frac{e^{x}}{2} \Rightarrow$

$$\Rightarrow C_2'(x) = -\frac{1}{2}e^{2x} - \frac{1}{2} \Rightarrow \boxed{C_2(x) = -\frac{1}{4}e^{2x} - \frac{1}{2}x.}$$

$$y(x) = y_0(x) + y_p(x) = c_1 \cdot y^{(1)}(x) + c_2\, y^{(2)}(x) +$$

$$+ c_1(x)\, y^{(1)}(x) + c_2(x)\, y^{(2)}(x) =$$

$$= (c_1 + c_1(x))\, y^{(1)}(x) + (c_2 + c_2(x))\, y^{(2)}(x) =$$

$$= \left(c_1 + \frac{1}{2}x - \frac{e^{-2x}}{4}\right)(e^{x}) \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \left(c_2 + \left(-\frac{1}{4}\right)e^{2x} - \frac{1}{2}x\right) \cdot$$

$$\cdot e^{-x} \cdot \begin{pmatrix} 1 \\ -1 \end{pmatrix} \cdot$$

$$\begin{pmatrix} y_1(x) \\ y_2(x) \end{pmatrix} = y(x) \Rightarrow y_1(x) = \left(C_1 + \frac{1}{2}x - \frac{e^{-2x}}{4}\right)e^x +$$

$$+ \left(C_2 - \frac{e^{2x}}{4} - \frac{x}{2}\right)e^{-x}$$

$$y_2(x) = \left(C_1 + \frac{x}{2} - \frac{e^{-2x}}{4}\right)e^x - \left(C_2 - \frac{e^{2x}}{4} - \frac{x}{2}\right)e^{-x}.$$

Apl. metoda variației c.

Exponențiala unei matrici

Fie $A \in \mathcal{M}_n(K)$, unde $K = \mathbb{R}$.

$\|A\| = \sqrt{\sum_{i=1}^{n} \sum_{j=1}^{n} |a_{ij}|^2}$ (normă)

$\|A - B\| = d(A, B)$, $\mathcal{M}_p(\mathbb{R}, d)$ = sp. metric normat

Considerăm <u>seria de puteri</u>: $\sum a_n x^n$ cu raza de conv. R

Dacă $\|A\| < R$, alegem $r > 0$ a.î. $\|A\| < r < R \Rightarrow$

$\Rightarrow$ seria numerică $\sum a_n \cdot r^n$ este abs. conv. (AC)

$\sum a_n \cdot A^n$, $\|a_n \cdot R^n\| = |a_n| \cdot \|A^n\| < |a_n| \cdot r^n =$

$= |a_n| \cdot r^n$

Cum $\sum |a_n \cdot \lambda^n|$ (c) $\Rightarrow \sum \| a_n A^n \|$ (c) $\Rightarrow$

$\Rightarrow \sum a_n A^n$ (A.C) deci (c) în $(\mathcal{M}_p(\mathbb{R}), d)$

$$\boxed{f(A) = \sum a_n A^n}$$

$e^x = \sum_{n \geq 0} \frac{x^n}{n!}$; $R = \infty$.

$\sum_{n \geq 0} \frac{1}{n!} \cdot A^n$ este (c) , $(\forall) A \in \mathcal{M}_p(\mathbb{R})$.

Notăm cu e^A = suma seriei = $\sum_{n\geq 0} \frac{1}{n!} A^n$

Proprietăți ale exponențialei unei matrici

Considerăm $A, B \in \mathcal{M}_n(\mathbb{R})$

1) $e^{0_n} = I_n$

2) $e^{\lambda I_n} = e^{\lambda} \cdot I_n$

3) $e^A \cdot e^B = e^{A+B}$, dacă $A \cdot B = B \cdot A$

4) $(e^A)^{-1} = e^{-A}$

5) $e^{T^{-1}AT} = T^{-1} e^A T$, $T \in \mathcal{M}_n(\mathbb{R})$, T = inversabilă

6) $e^{\begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}} = \begin{pmatrix} e^{\lambda_1} & & 0 \\ & \ddots & \\ 0 & & e^{\lambda_n} \end{pmatrix}$

7) $\left(e^{tA}\right)' = A \cdot e^{tA}$

⊛ $(0_n)^0 = I_n$

Dem.: 1) $e^{O_n} = \sum_{k\geq 0} \frac{1}{k!} \cdot O_n^k = \frac{1}{0!} I_n + \underbrace{\sum_{k\geq 1} \frac{1}{k!} \cdot O_n^k}_{O_n} = I_n$

2) $e^{\lambda I_n} = \sum_{k\geq 0} \frac{1}{k!} (\lambda I_n)^k = \sum_{k\geq 0} \frac{1}{k!} \lambda^k \cdot \underbrace{O_n^k}_{I_n} \cdot I_n =$

$= \left(\sum_k \frac{1}{k!} \cdot \lambda^k \right) I_n = e^{\lambda} I_n .$

3) $e^A \cdot e^B = \left(\sum_{K\geq 0} \frac{1}{K!} A^K\right)\left(\sum_{P\geq 0} \frac{1}{P!} B^P\right) \Rightarrow$

notăm $\boxed{K+P=Q}$

$$\Rightarrow \sum_{Q\geq 0}\left(\sum_{K+P=Q} \frac{1}{K!} A^K \cdot \frac{1}{P!} B^P\right) =$$

$$= \sum_{Q\geq 0}\left(\sum_{K=0}^{Q} \frac{1}{K!} A^K \cdot \frac{1}{(Q-K)!} \cdot B^{Q-K}\right) =$$

$$= \sum_{Q\geq 0}\left(\frac{1}{Q!} \sum_{K=0}^{Q} \frac{Q!}{K!(Q-K)!} A^K \cdot B^{Q-K}\right)$$

$$= \sum_{q \geq 0} \left(\frac{1}{q!}\right)\left(\sum_{k=0}^{q} C_q^k A^k \cdot B^{q-k}\right) = \sum_{q \geq 0} \frac{1}{q!}(A+B)^q = e^{A+B}$$

4) $e^A \cdot e^{-A} = e^{0_n} = I_n \Rightarrow (e^A)^{-1} = e^{-A}$.

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5) $e^{T^{-1}AT} = \sum_{k\geq 0} \frac{1}{k!} (T^{-1}AT)^k = \frac{1}{0!} I_n + \sum_{k\geq 1} \frac{1}{k!} \underbrace{T^{-1}AT \cdot T^{-1}AT \cdot \ldots \cdot T^{-1}AT}_{\text{de ori}}$

$$= \frac{1}{0!} I_n + \sum_{k\geq 1} \frac{1}{k!} T^{-1}A^k T = \frac{1}{0!} T^{-1} I_n T + \sum_{k\geq 1} \frac{1}{k!} T^{-1} A^k T =$$

$$= T^{-1} \left(\sum_{k\geq 0} \frac{1}{k} A^k \right) \cdot T = T^{-1} \cdot e^A \cdot T$$

6) $e^{\begin{pmatrix} \lambda_1 & \ldots & 0 \\ 0 & \ldots & \lambda_n \end{pmatrix}} = \sum_{k\geq 0} \frac{1}{k!} \begin{pmatrix} \lambda_1 & \ldots & 0 \\ 0 & \ldots & \lambda_n \end{pmatrix}^k = \sum_{k\geq 0} \frac{1}{k!} \begin{pmatrix} \lambda_1^k & \ldots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \ldots & \lambda_n^k \end{pmatrix} =$

$$= \begin{pmatrix} \sum_{k\geq 0} \frac{1}{k!} \lambda_1^k & \ldots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \ldots & \sum_{k\geq 0} \frac{1}{k!} \lambda_n^k \end{pmatrix} = \begin{pmatrix} e^{\lambda_1} & \ldots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \ldots & e^{\lambda_n} \end{pmatrix}.$$